

Automated Network Request Management in ServiceNow

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Project Name	Automated Network Request Management in ServiceNow

This document covers the complete end-to-end build procedure for the Automated Network Request Management system. It follows a natural, step-by-step creation flow from database schema design to frontend portal configuration, low-code workflow automation, and final testing verification. Each stage details the exact setup instructions along with the specific screenshots required for project evidence submission.

Stage 1: Setting Up the Custom Table (u_database_table)

Before building forms or workflows, I needed a dedicated place to store every network request cleanly without cluttering default tables like Incident. For this, I created a custom table called Database Tables (u_database_table).

I navigated to All > System Definition > Tables and clicked New. In the Table Label field, I entered Database Tables, and ServiceNow automatically generated the name u_database_table. To ensure every request receives a unique tracking identifier, I navigated to the Controls tab, enabled Auto-number, set the prefix to DBT, and selected 7 digits. This automatically formats records as NETDB0001009.

In the Columns tab, I added the fields required to hold form inputs and track request progress:

- **Requester Profile:** Requested For (Reference to sys_user), Opened on behalf of (Reference to sys_user), and User Name (String).
- **Contact Information:** Phone Number (String), Mobile Number (String), and Email ID (String) for sending automatic notifications.
- **Network Details:** Type of Connection (Choice: New, Existing) and Existing ID (String) to capture existing port or line numbers.
- **Billing & Location:** Total Amount (Decimal), Mode of Payment (Choice: UPI, Card), and Address (Multi-line String, 4000 chars) for workstation delivery.
- **Lifecycle State:** State (Choice: Awaiting Approval, Approved, Rejected) to track the request throughout the workflow.

I saved the table and verified that default Access Control Lists (ACLs) were created, granting read access to requesters and full control to administrators.

Screenshot 1: Custom Table & Auto-Number Setup

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* LabelDatabase Tables

* Nameu_database_tables

ApplicationGlobal

Remote Table

Gender

ColumnsControlsApplication Access

Table Columnsfor textSearch

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Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)		32	false
Updated	Date/Time	(empty)		40	false
✕ Mobile Number	String	(empty)		40	false
✕ Database Number	String	(empty)		40	false
✕ Customer Address	String	(empty)		40	false
Created by	String	(empty)		40	false
✕ Mode of Payment	String	(empty)		40	false
✕ Total Amount	String	(empty)		40	false
✕ Approval Status	String	(empty)		40	false
Created	Date/Time	(empty)		40	false
✕ Type of Connection	String	(empty)		40	false
✕ Requested For	String	(empty)		40	false
✕ Number	String	(empty)		40javascript:global.getNextObjNumberPadded();	false
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
✕ Assignment to	Reference	User		32	false
+ Insert a new row...					

ColumnsControlsApplication Access

Extensible

Live feed

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

PrefixNETDB

Number1,000

Number of digits7

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls

* User roleu_database_tables_user

Stage 2: Building the Service Catalog Item & Dynamic Form Rules

With the database table in place, I configured the user-facing request item in the Service Catalog. To keep the form intuitive and avoid human error, I added dynamic rules so conditional fields only show up when relevant.

I navigated to Service Catalog > Catalog Definitions > Maintain Items and clicked New. I titled the item Network Request and categorized it under Network request Management. In the Variables tab, I created all 7 variables matching the custom table fields

To handle conditional visibility without writing custom client scripts, I opened the Catalog UI Policies tab and clicked New. I set the condition to Type of Connection is Existing, leaving On Load = True and Reverse if false = True. Under Catalog UI Policy Actions, I targeted existing_id and set Make Visible = True and Make Mandatory = True. This keeps the Existing ID field hidden for new connections while enforcing it whenever an existing connection is selected.

Screenshot 2: Service Catalog Item Configuration

servicenow

AllFavoritesHistoryWorkspacesAdmin

Catalog Item - Network Request

Search

CopyTry ItUpdateEdit in Catalog BuilderDelete

Catalog Item Network Request

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

NameNetwork Request

CatalogsService Catalog

CategoryNetwork Standard Changes

State-- None --

Checked out-- None --

OwnerSystem Administrator

Secondary owners

ApplicationGlobal

Active

Fulfillment automation levelUnspecified

Item DetailsProcess EnginePicturePricingPortal Settings

Short descriptionNetwork request Management

Description

Variables (7)Variable Sets (1)Catalog UI Policies (1)Catalog Client Scripts (1)Available ForNot Available ForCategories (1)Catalogs (1)Catalog Data Lookup DefinitionsRelated ArticlesRelated Catalog Items

Assigned Topics

OrderSearch

Actions on selected rows...New

Catalog Item = Network Request

Type	Question	Order
Reference	Requested For	100
Single Line Text	Mobile Number	200
Multiple Choice	Type of Connection	300
Single Line Text	Enter your existing id	400
Single Line Text	Total Amount	500
Multiple Choice	Mode of Payment	600
Multi Line Text	Address	700

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Screenshot 3: Dynamic Catalog UI Policy Rule

Variables (7)
Variable Sets (1)
Catalog UI Policies (1)
Catalog Client Scripts (1)
Available For
Not Available For
Categories (1)
Catalogs (1)
Catalog Data Lookup Definitions
Related Articles
Related Catalog Items

Assigned Topics

for text
Search
Actions on selected rows...
New

Catalog item = Network Request

	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	Display Existing ID when Type of Connection is Existing	(empty)		true	true	false	2026-09-30 02:45:43	100

<< 1 to 1 of 1 >>

Catalog UI policies are similar to standard UI policies. Catalog UI policies dynamically change variables that are part of a catalog item or change how variable sets are handled. Policies can also be applied when the variables are present in a Requested Item or Catalog Task form. [More Info](#)

Applies to: A Catalog Item
Application: Global
Active: ☒

* Catalog item: Network Request
* Short description: Display Existing ID when Type of Connection is Existing

When to Apply
Script

Catalog UI policy actions are applied only if all the following conditions are met:
1. The catalog UI policy is **Active**
2. The items in the **Conditions** field evaluate to true
3. The field specified in the catalog UI policy is present on the specified catalog item

Catalog Conditions
Add Filter Condition
Add OR Clause

type_of_connection is Existing
AND OR

Applies on a Catalog Item view ☒
Applies on Catalog Tasks ☐
Applies on Requested Items ☐

Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form

On load ☒

Reverse the effects of the catalog UI policy actions when the Conditions evaluate to false

Reverse If false ☒

Stage 3: Testing the Service Portal Intake (/sp)

To make sure everything worked as expected from an end-user standpoint, I opened the ServiceNow Service Portal (/sp) and performed a live test submission.

I opened the portal at /sp, typed Network Request into the search bar, and opened the catalog item. I verified that my contact details were loaded properly, then tested the UI policy by toggling the connection type: clicking Existing immediately revealed the mandatory Existing ID field without refreshing the page. I filled in the required values, picked UPI as the payment mode, entered the total fee, typed the delivery address, attached a verification file, and clicked Submit.

Screenshot 4: Service Portal Catalog Item Search

Home > Search

network request

Sources

All
Knowledge Bases
Catalogs

FILTERS

Knowledge Bases

Clear All

Category ▶

Apply

Author ▶
Last modified ▶
View Count ▶

Catalogs

Clear All

Category ▶

Apply

All results for "network request"


Network Request

Network request Management


Request for Backup

Backup system, data files or database


Apple iPhone 13

Request for Apple iPhone 13
\$799.00


Apple iPhone 13 pro

Request for Apple iPhone 13 pro
\$999.00 + \$45.00 Monthly

Screenshot 5: Network Request Portal Form (Initial Intake View)

Network Request

Network request Management

* Indicates required

Requested For

System Administrator

×

▼

Opened on behalf of

▼

User Name

Phone Number

Email ID

Mobile Number

* Type of Connection

☒ New

☐ Existing

Total Amount

Mode of Payment

☒ UPI

☐ Card

Address

Add attachments

↑

[Choose a file](#) or drag it here.
Copy and paste clipboard files here.

Screenshot 6: Dynamic UI Policy in Action (Existing ID Revealed)

Network Request

Network request Management

* Indicates required

Requested For

X ▼

Opened on behalf of

X ▼

User Name

Phone Number

Email ID

Mobile Number

* Type of Connection

☐ New

☒ Existing

Enter your existing id

Total Amount

Mode of Payment

☒ UPI

☐ Card

Address

Add attachments


[Choose a file](#) or drag it here.
Copy and paste clipboard files here.

Screenshot 7: Request Submission Confirmation

The screenshot shows the ServiceNow interface for a request submission confirmation. The top navigation bar includes links for Knowledge, Catalog, Requests, Approvals (with a red notification icon), System Status, Cart, Tours, and a user profile for System Administrator. The main header shows 'Home > Request Summary - REQ0010009' and a search bar labeled 'Search Catalog'. Below this, a summary box displays: Submitted: 2026-10-06 22:36:07, Request Number: REQ0010009, and Estimated Delivery: 2026-10-06. A table lists the request items:

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Network Request	2026-10-06		---	1	---

The total at the bottom right is \$0.00. There are icons for a mobile app and a chat bubble in the bottom left and right corners, respectively.

Stage 4: Building the 10-Step Flow Designer Workflow

Once the Service Portal form was ready, I automated the backend processing in Flow Designer without writing scripts. The 10-step flow captures form variables, inserts a custom record, handles manager approvals, and routes conditional outcomes.

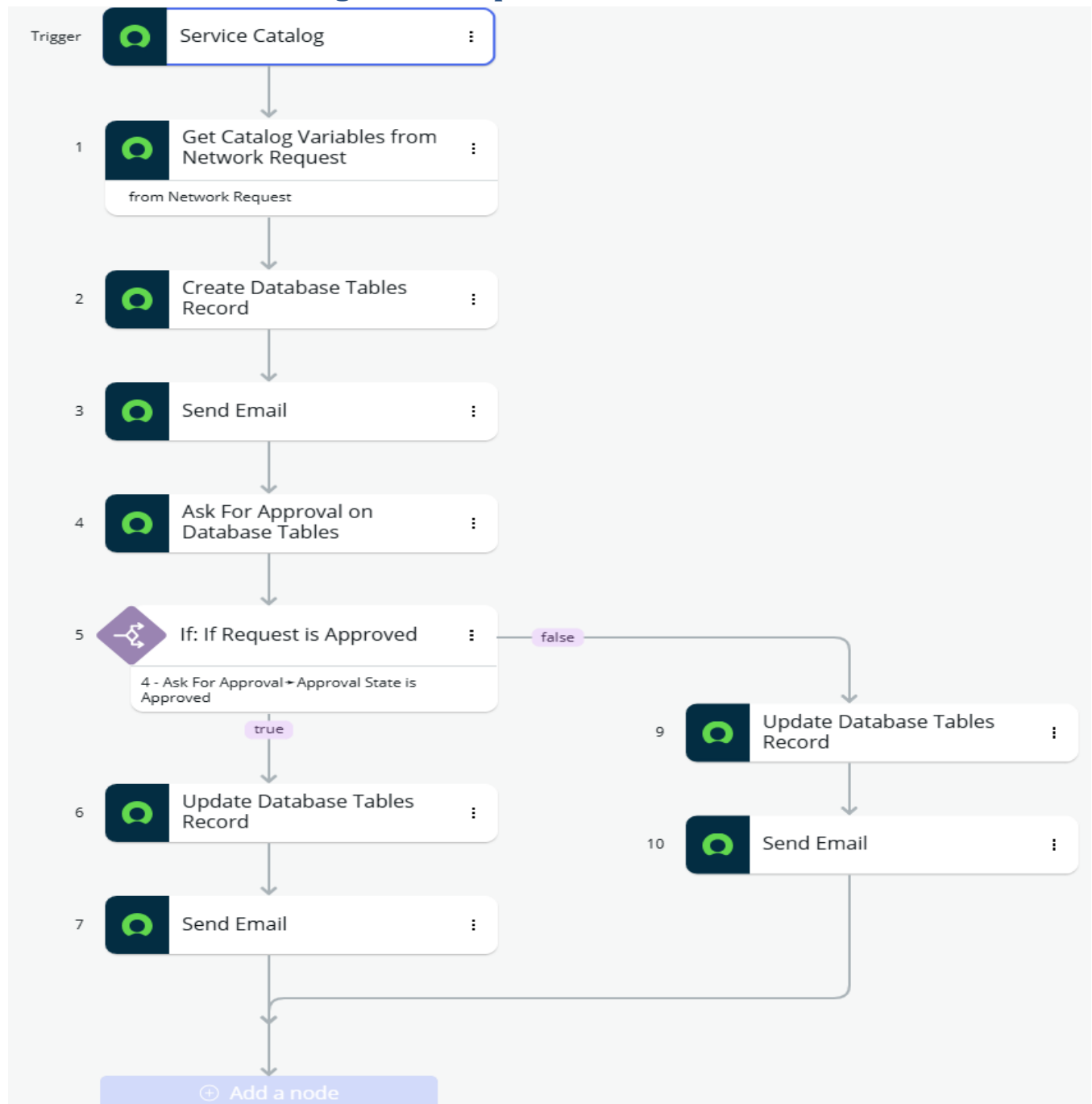
- **Trigger: Service Catalog** — Starts automatically upon catalog item submission.
- **Step 1: Get Catalog Variables from Network Request** — Loads submitted form values into data pills.
- **Step 2: Create Database Tables Record** — Inserts a record into Database Tables (u_database_table) with the captured variables.
- **Step 3: Send Email** — Sends an immediate confirmation email with the tracking reference to the user.
- **Step 4: Ask For Approval on Database Tables** — Generates an approval task and assigns it to the manager.
- **Step 5: If: If Request is Approved** — Checks if the approval state is *Approved*.
 - **Approved:**
 - **Step 6: Update Database Tables Record** — Sets record state to *Approved*.
 - **Step 7: Send Email** — Sends an approval confirmation email to the user.

➤ **Rejected:**

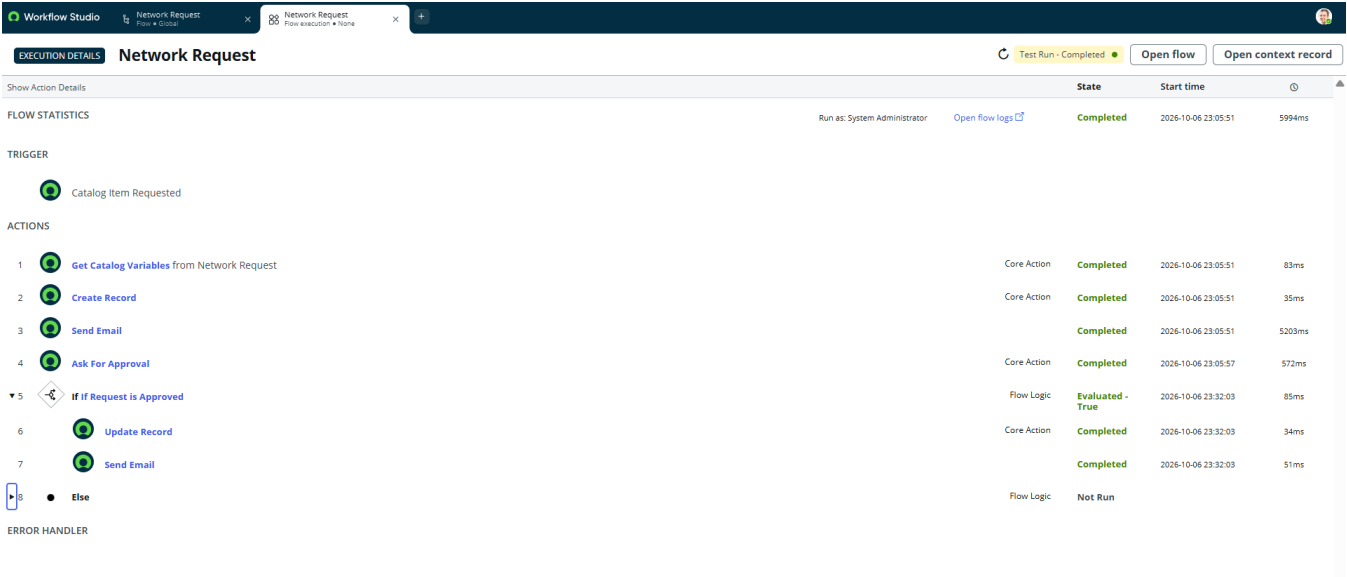
- **Step 9: Update Database Tables Record** — Sets record state to *Rejected*.
- **Step 10: Send Email** — Sends a rejection notice email to the user.

After verifying the logic with **Test Flow**, I clicked **Activate** so every new portal submission executes this automation automatically.

Screenshot 8: Flow Designer 10-Step Workflow Canvas



Screenshot 9: Flow Execution Details (Runtime Traces)



Stage 5: Data Persistence, Governance & Notification Verification

The final stage was verifying that submitted tickets persist properly in our custom table, integrate with platform approvals, and trigger outbound notifications.

Here is how I verified each component:

- 1. Database Record Check:** In database list verified that submitting the portal form created a row with auto-generated ID NETDB0001009 and initial state **Awaiting Approval**, correctly saving all user, payment, and address values.
- 2. Approval Gate Execution:** In sysapproval_approver_list.do, verified that Step 4 created an approval task for the manager. Approving it immediately shifted the database record state to **Approved** via the flow's true branch.
- 3. Email Delivery Audit:** In email list, confirmed two outgoing emails were sent: the initial submission receipt with the tracking ID (Step 3) and the approval notice (Step 7).
- 4. Security ACL Verification:** Verified table ACLs by impersonating an end user, confirming requesters can only view their own submissions while admins retain full write access.

Screenshot 10: Database Tables Record View (u_database_table)

servicenow All Favorites History Workspaces Admin Database Tables ☆										
Database Tables Number Search Actions on selected rows... New										
All										
	Number	Approval Status	Assignment to	Customer Address	Database Number	Mobile Number	Mode of Payment	Requested For	Total Amount	
☐	NETDB0001019	Approved	ITIL User	WIZ street	RITM0010008	78456133466	UPI	6816f79cc0a8016401c5a33be04be441		
	NETDB0001008	Requested	(empty)	maka maak	RITM0010004		UPI			
	NETDB0001016	Approved	ITIL User	XYZ street	RITM0010007		UPI			
	NETDB0001015	Rejected	Abraham Lincoln	ABC Street	RITM0010006	9785612312	UPI		10	
	NETDB0001021	Approved	ITIL User	ABC street, 321092	RITM0010009	9856124536	UPI	6816f79cc0a8016401c5a33be04be441	100	
	NETDB0001017	Approved	ITIL User	XYZ street	RITM0010007		UPI			
	NETDB0001014	Approved	ITIL User	ABC Street	RITM0010006	9785612312	UPI		10	
	NETDB0001012	(empty)								
	NETDB0001020	Requested	(empty)	ABC street, 321092	RITM0010009	9856124536	UPI	6816f79cc0a8016401c5a33be04be441	100	
☒	NETDB0001013	Approved	ITIL User	ABC Street	RITM0010006	9785612312	UPI		10	
	NETDB0001018	Requested	ITIL User	WIZ street	RITM0010008	78456133466	UPI	6816f79cc0a8016401c5a33be04be441		
	NETDB0001009	Approved	ITIL User	123 ABC Street	RITM0010005		UPI			

Screenshot 11: Approval Gate Record (sysapproval_approver)

servicenow All Favorites History Workspaces Admin Approvals ☆							
Approvals Approval for Search Actions on selected rows...							
All > Sys ID = NULL.or. Approver = System Administrator							
	Approval for	Approver	State	Comments	Created	Created by	Approving
☐	(empty)	System Administrator	Approved		2026-10-06 23:05:57	admin	Database Tables: NETDB0001021
	(empty)	System Administrator	Approved		2026-09-30 23:55:40	system	Database Tables: NETDB0001009
	(empty)	System Administrator	Approved		2026-10-02 08:18:02	system	Database Tables: NETDB0001018
	(empty)	System Administrator	Approved		2026-10-02 08:18:59	admin	Database Tables: NETDB0001019
	(empty)	System Administrator	Rejected	2026-10-02 00:27:34 - System Administrat	2026-10-02 00:26:52	admin	Database Tables: NETDB0001015
	(empty)	System Administrator	Approved		2026-10-02 00:12:53	system	Database Tables: NETDB0001013
	(empty)	System Administrator	Approved		2026-10-02 01:56:46	admin	Database Tables: NETDB0001017
	(empty)	System Administrator	Approved		2026-10-02 01:51:30	system	Database Tables: NETDB0001016
	(empty)	System Administrator	Rejected	2026-10-02 00:21:36 - System Administrat	2026-10-02 00:16:22	admin	Database Tables: NETDB0001014
	(empty)	System Administrator	Approved		2026-10-06 22:36:09	system	Database Tables: NETDB0001020

Screenshot 12: Outbound Email Notification Logs (sys_email)

servicenow

AllFavoritesHistoryWorkspacesAdmin

Emails

Search

Actions on selected rows...New

All > Created on Today

Created	Recipients	Subject	Type	Notification type	User ID
2026-10-06 00:00:06	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-10-06 01:00:07	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-10-06 01:00:08	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-10-06 23:32:03	abraham.lincoln@example.com	Your Network Request Has Been Approved	send-ready	SMTP	(empty)
2026-10-06 01:00:06	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-10-06 23:05:57	abraham.lincoln@example.com	Database Record Is Created	send-ready	SMTP	(empty)
2026-10-06 22:36:15	admin@example.com	Request REQ0010009 was created	send-ready	SMTP	(empty)
2026-10-06 22:36:15	admin@example.com	Request REQ0010009 was approved	send-ready	SMTP	(empty)
2026-10-06 23:47:20	abraham.lincoln@example.com	Your Network Request Has Been Approved	send-ready	SMTP	(empty)
2026-10-06 10:01:15	admin@example.com	[ITSM Analytics] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-10-06 22:36:09	abraham.lincoln@example.com	Database Record Is Created	send-ready	SMTP	(empty)

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Preview Email

Hello Abraham Lincoln

Your network request has been approved and your account setup/update is now complete.

Thank you!

Ref:MSG0002034_oh6YH6kgezI35oiB8mJJ

Close

Screenshot 13: Table Access Control List (ACL) Rules

Access Controls (4)Security Data FiltersLabels (1)Database Indexes (2)Table Subscription Configuration (1)

NameSearch

Actions on selected rows...

Access Controls

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_database_tables	Allow If	read	record	true	admin	2026-09-29 20:55:34
u_database_tables	Allow If	delete	record	true	admin	2026-09-29 20:55:34
u_database_tables	Allow If	create	record	true	admin	2026-09-29 20:55:33
u_database_tables	Allow If	write	record	true	admin	2026-09-29 20:55:34

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